
CHANGEscout: NLP BASED PRIORITIZATION OF INFRASTRUCTURE CHANGE SOURCES FOR TOPOGRAPHIC DATA PRODUCTION

Lorenz Joss

Federal Office of Topography swisstopo
Wabern, Switzerland
lorenz.joss@swisstopo.ch

Final project report
CAS Natural Language Processing 2025/2026
University of Bern
May 11, 2026

ABSTRACT

Public authorities publish information about planned, ongoing, and completed infrastructure projects on official web pages. For topographic data production, these texts can contain useful signals about changes that may require human review, but manually screening such sources is time consuming and difficult to reproduce. This project investigates whether natural language processing methods can prioritize official cantonal web documents so that potentially relevant infrastructure changes can be reviewed more efficiently. ChangeScout is implemented as a human in the loop NLP workflow and evaluated on a manually reviewed annotation dataset. The project compares rule based thematic scoring, TF IDF Logistic Regression, LLM based triage, hybrid candidate selection, and LLM based explanation support. The results show that deterministic scoring provides a useful and transparent baseline, while TF IDF Logistic Regression provides the strongest recall for actionable review selection among the individual methods. LLM based methods are less suitable as standalone prioritization methods because of their conservative recall behavior, but they add value for explanation and review support. The recommended use of ChangeScout is therefore not automatic updating of topographic data, but reproducible prioritization of official sources for human review.

Keywords natural language processing · information retrieval · lead prioritization · human in the loop · topographic data production · official web sources

1 Introduction

The Federal Office of Topography swisstopo produces and maintains official geodata for Switzerland. One of its core vector data products is swissTLM3D, the large scale topographic landscape model of Switzerland [1]. It is used as a basis for national mapping and many other geodata products. The model contains objects that shape the landscape, including roads, paths, buildings, water bodies, ground cover, engineering structures, and other topographic features. Its production and updating are mainly based on current aerial images and photogrammetric interpretation. This provides systematic national coverage, but it also means that real world changes are often incorporated only after they are visible in imagery and after the production cycle reaches the affected area. For infrastructure objects such as roads, junctions, bridges, tunnels, or separated cycling paths, the reaction time should be shorter whenever possible. Additional information sources are therefore relevant for production, especially where no complete notification system or structured data exchange exists.

Infrastructure changes are announced in procurement platforms, project pages, planning documents, media releases, and official websites of cantons, municipalities, and federal agencies. These sources can contain early indications of works that may later become visible in the landscape and may require updates to topographic data. However, they are written for administrative, political, procurement, or public communication purposes rather than as structured geodata change

messages. They are heterogeneous, text heavy, distributed across different platforms, and often mix relevant geometric information with maintenance, temporary traffic management, planning context, or general project descriptions. This project uses manually curated official cantonal web pages as a controlled source class for evaluating the NLP problem. The corresponding source registries are stored as versioned YAML files in the project repository [2]. This source choice is a scope decision for the project. It does not imply that cantonal websites are the only or best source for a future operational system. Procurement data, municipal sources, and structured exchanges may be equally or more relevant, but they introduce additional acquisition challenges such as PDF extraction, APIs, dynamic websites, or source specific parsing. This report therefore focuses on the downstream NLP question once official text sources have been collected and normalized.

The central question of this project is how well rule based methods, classical NLP, and LLM based methods can prioritize official web texts so that TLM relevant infrastructure changes can be reviewed more efficiently by humans. In this report, selected review candidates are called leads. The goal is not to automatically update swissTLM3D. Final production judgement remains with domain experts. A lead is treated as a document that should be reviewed, not as a confirmed topographic change. More specifically, the project evaluates whether deterministic thematic scoring can identify potentially relevant sources with useful recall, whether a classical TF IDF Logistic Regression baseline improves over deterministic scoring, whether LLM based methods can improve triage or explanation quality without being used as hard exclusion signals, and which hybrid strategy provides the best tradeoff between recall, precision, false negatives, and manual review workload. The project contributes a reproducible NLP workflow for official cantonal web sources, a manually reviewed annotation dataset, a comparison of deterministic, classical, and LLM based methods, and an evaluation of hybrid lead selection strategies. The final output is a prioritized review queue for topographic data production, not an automatic change detection system in the geographic sense.

2 Data and Annotation

The dataset used in this project was built from official web pages that were transformed into normalized text documents and then manually reviewed. The purpose was to create a controlled evaluation basis for the NLP task. The implementation, configuration files, and evaluation artifacts are provided in the project repository [3]. The data preparation therefore balances two requirements: the sources should reflect realistic public information that can be relevant for topographic data production, while the collection and annotation process must remain explicit, reproducible, and suitable for method comparison.

2.1 Source Scope

The project uses manually curated official cantonal web pages as source material. Cantonal project and infrastructure pages were selected because they often contain official descriptions of planned, ongoing, or completed works and because several of these pages are technically accessible enough for a controlled experiment. The corresponding source registries are stored as versioned YAML files in the project repository and define the official entry pages and URL inclusion patterns used for discovery [2].

For the NLP methods, the unit of analysis is one crawled and cleaned web document. The relevant input consists of the extracted document title and the cleaned body text. The evaluation corpus contains documents from multiple cantons and source types to include different official communication styles. This makes the dataset broader than a single source experiment, while still keeping the collection process explicit and reproducible. This source scope is deliberately limited. The project does not claim to cover all relevant public infrastructure information. Municipal websites, procurement platforms, PDF documents, APIs, and structured data exchanges may be equally or more relevant for a future operational system, but they introduce additional acquisition challenges and were not systematically covered in this report.

2.2 Data Flow and Preprocessing

The processing workflow transforms curated web sources into normalized text documents and then into evaluation datasets. Figure 1 shows the simplified data flow. Discovery starts from configured source registries, extracts candidate URLs, and keeps URLs matching the configured inclusion patterns. Crawling then fetches these pages and stores raw HTML together with metadata such as source identifier, URL, fetch timestamp, HTTP status, and content hash. These steps define which documents enter the corpus and make it possible to inspect the original HTML later if cleaning or annotation decisions are unclear.

HTML cleaning extracts the title and main body text, removes obvious boilerplate, normalizes whitespace, applies language filtering, and enforces a minimum text length. These are technical quality steps, not semantic relevance

decisions. A later hard off topic filter removes only clearly unrelated documents and is deliberately conservative. Weak or ambiguous infrastructure texts are preserved for scoring, annotation, classification, or human review. The current workflow is reproducible for the curated and similarly structured source registries, but it is not a general solution for all official websites. Dynamic pages, PDF heavy sources, APIs, and source specific layouts may require additional extraction logic. This limitation affects source acquisition, but it is separate from the main NLP evaluation in this report.

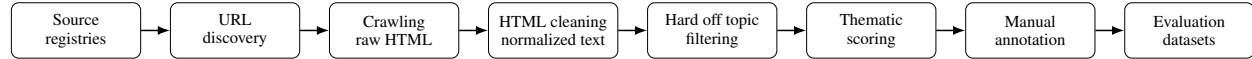


Figure 1: Simplified data flow from curated source registries to evaluation datasets.

2.3 Annotation Schema

Manual annotation was performed on the cleaned document title and body text. The label does not express whether a construction project is important in a general sense. It expresses whether the source text gives sufficient evidence that a persistent TLM geometry update would be required. The core annotation question is whether a mapper would need to add, remove, split, connect, disconnect, reshape, or redraw a TLM geometry based on the information in the source [4].

This distinction is central for the project. General construction activity is not automatically relevant for TLM. Road resurfacing, temporary traffic management, road markings, drainage work, administrative information, or public transport changes are not relevant geometry changes by themselves. They are labelled as not relevant unless the text also implies a persistent geometry update. Conversely, changes such as new roads, changed alignments, new junctions, new roundabouts, new underpasses, physically separated cycling paths, or mapped traffic islands can be relevant when the source text provides enough evidence. Some cases remain ambiguous. For example, a road widening can be irrelevant if it stays on the same axis, but relevant if it creates a separated carriageway, a new access, or a changed alignment.

The field `change_type` was used to describe the type of potential TLM effect. `topology` refers to changes in the road or path network, such as new connections, junctions, roundabouts, entries, exits, or changed axes. `geometry` refers to mapped geometry changes without a clear network connectivity effect, for example a mapped traffic island. `attribute_only` refers to changes that affect only attributes and not mapped geometry. `none` is used when no relevant TLM update is expected. Attribute only cases were retained in the annotation schema for completeness, but they were not developed into a separate modelling target because they were sparse and not the central question of this project.

Table 1: Annotation schema used for the manually reviewed documents.

Field	Meaning
<code>tlm_relevant</code>	Sufficient textual evidence for a persistent TLM geometry update.
<code>review_required</code>	Plausible TLM signal, but the source text does not provide enough evidence for a confirmed decision.
<code>change_type</code>	Type of expected effect: <code>topology</code> , <code>geometry</code> , <code>attribute_only</code> , or <code>none</code> .
<code>notes</code>	Short explanation of the annotation decision.
<code>trriage_class</code>	Derived evaluation class based on <code>tlm_relevant</code> and <code>review_required</code> .

The manually assigned fields are `tlm_relevant`, `review_required`, `change_type`, and `notes`. The field `trriage_class` is derived afterwards from the two primary decision labels and is used as the main class variable for evaluation. The combination `tlm_relevant = true` and `review_required = true` is invalid because a document cannot be both confirmed relevant and still require review due to insufficient evidence [4].

Table 2: Derivation of the triage class from the primary annotation labels.

<code>tlm_relevant</code>	<code>review_required</code>	<code>trriage_class</code>
true	false	<code>confirmed_relevant</code>
false	true	<code>needs_review</code>
false	false	<code>not_relevant</code>
true	true	<code>invalid</code>

The derived class `confirmed_relevant` is used when the source text contains sufficient evidence for a persistent geometry update. The class `needs_review` is used when the text contains a plausible signal, but the source itself does not provide enough information for a final binary decision. In such cases, a reviewer would need to inspect additional information, such as plans, project documents, maps, or later project updates. The class `not_relevant` is used when no plausible TLM geometry update signal is present. This three class structure is important because ambiguous but review worthy documents are not forced into the same category as clearly irrelevant documents.

2.4 Evaluation Datasets

The frozen annotation dataset contains 348 manually reviewed sources. Table 3 shows the class distribution. The dataset includes confirmed relevant cases, review cases, and not relevant cases. This mix is necessary because the practical task is not only to identify confirmed geometry evidence, but also to preserve documents that are uncertain yet useful for human inspection.

Table 3: Class distribution of the frozen annotation dataset.

Triage class	Documents
<code>confirmed_relevant</code>	124
<code>needs_review</code>	84
<code>not_relevant</code>	140
Total	348

Three evaluation datasets were derived from the frozen annotation dataset. They represent different evaluation questions rather than competing versions of the same label. The `strict_binary` dataset evaluates confirmed TLM relevance. It excludes `needs_review` cases because these cases do not provide enough textual evidence for a confirmed relevance decision. Including them as either positive or negative would mix confirmed geometry evidence with unresolved review uncertainty.

The `actionable_binary` dataset evaluates review usefulness. Here, both `confirmed_relevant` and `needs_review` are positive because both should enter a human review queue. This task is closer to the practical goal of ChangeScout, where missing a review worthy source can be more costly than showing an additional false positive. The `triage_3class` dataset keeps all three classes and evaluates whether a method can separate confirmed relevance, review worthy uncertainty, and non relevant sources.

Table 4: Evaluation datasets derived from the frozen annotation dataset.

Dataset	Rows	Train	Test	Target definition
<code>strict_binary</code>	264	211	53	Positive: <code>confirmed_relevant</code> . Negative: <code>not_relevant</code> , <code>needs_review</code> excluded.
<code>actionable_binary</code>	348	278	70	Positive: <code>confirmed_relevant</code> and <code>needs_review</code> . Negative: <code>not_relevant</code> .
<code>triage_3class</code>	348	278	70	Three class target with <code>confirmed_relevant</code> , <code>needs_review</code> , and <code>not_relevant</code> .

All method comparisons in this report use these frozen datasets and fixed splits. This keeps the evaluation reproducible and avoids mixing different task definitions. Confirmed relevance, actionable review selection, and three class triage are evaluated separately because they answer different practical questions.

3 Exploratory Data Analysis

The exploratory data analysis describes the annotated corpus in terms of document length, source composition, and thematic score distribution. These characteristics provide context for the later comparison of ranking and classification methods.

3.1 Document Length

The documents in the dataset differ substantially in length. Figure 2 shows median, mean, and interquartile range of the word count by triage class. Confirmed relevant documents are longer on average than the other two classes. Their median length is 357 words, compared with 264 words for needs_review and 299 words for not_relevant. This suggests that confirmed project descriptions often provide more detailed measure information. At the same time, the distributions overlap strongly, and source type differences cannot be reduced to document length alone. Document length is therefore not a meaningful relevance signal by itself, but it may still influence NLP methods because longer texts provide more opportunities for relevant keywords or geometric evidence to appear.

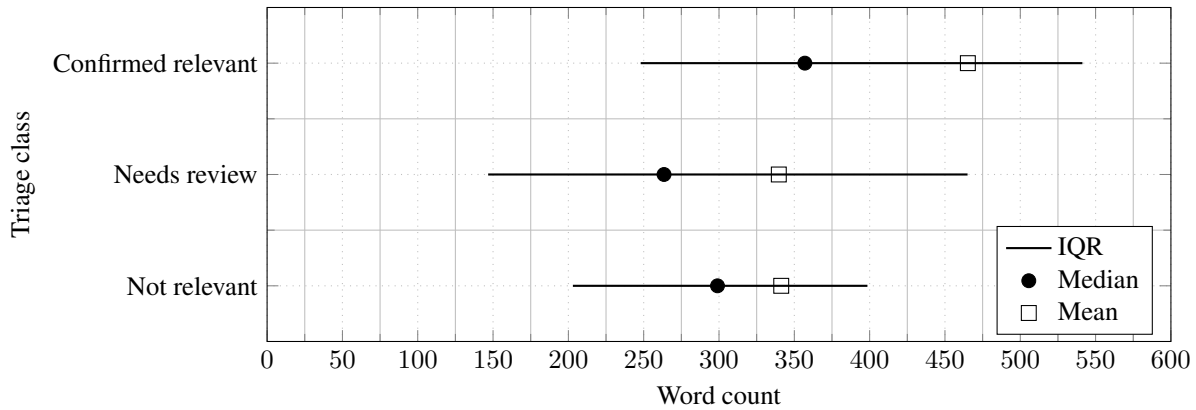


Figure 2: Document length distribution by triage class. The horizontal line shows the interquartile range, the filled circle shows the median, and the open square shows the mean. Confirmed relevant documents tend to be longer, but the overlap between classes remains substantial.

3.2 Source Composition

The annotation dataset contains documents from nine cantons. Table 5 shows the class distribution by canton prefix derived from the `source_id`. The distribution is not balanced by canton. Some cantons contribute mostly relevant project pages, while others contribute more hard negatives or broader public communication pages. For example, Solothurn contributes many `not_relevant` documents, while Bern and Zurich contain a high share of `confirmed_relevant` cases. This source composition is useful for method comparison because it introduces different official communication styles, but it also means that the dataset should not be interpreted as a representative sample of all Swiss public infrastructure information.

Table 5: Class distribution by canton prefix derived from `source_id`.

Canton	confirmed_relevant	needs_review	not_relevant	Total
AG	11	1	12	24
BE	24	8	4	36
BS	11	20	18	49
GR	2	1	1	4
LU	8	12	1	21
SG	21	18	32	71
SO	5	7	60	72
ZG	18	9	9	36
ZH	24	8	3	35
Total	124	84	140	348

The source composition is not only geographic, but also textual. Most documents are cantonal project pages, while 70 documents are media releases from the Solothurn media sample. This media sample was included deliberately to add hard negatives to the dataset. Media releases are official and canton specific, but they cover many different topics, including government decisions, schools, safety, administration, public notices, events, infrastructure, and occasionally

TLM relevant changes. They are therefore a noisy source type: relevant infrastructure signals are rare and mixed with many unrelated texts. This makes them useful for testing whether a method can prioritize documents with stronger geometric evidence instead of overrating broadly written official communication. Project pages, in contrast, are usually more focused on a specific infrastructure object or project.

Table 6: Source type composition, text length statistics, and manually annotated class distribution.

Source type	Docs	Median words	Mean words	confirmed	review	not relevant
Project pages	278	308.0	398.1	120	78	80
Media releases	70	317.5	333.4	4	6	60
Total	348			124	84	140

The text length statistics show that media releases are not simply shorter documents. Their median word count is similar to the project page median. The main difference is the manually annotated class distribution: 60 of 70 media releases are labelled `not_relevant`, while project pages contain a much higher share of `confirmed_relevant` and `needs_review` cases. The counts in Table 6 therefore describe the ground truth composition of the dataset, not model predictions. This confirms that source type affects the noise level of the dataset and creates a more realistic evaluation setting, because a review support system must avoid overprioritizing broadly written official communication texts only because they come from official sources or occasionally mention infrastructure.

3.3 Thematic Score Distribution

The deterministic thematic score already shows a clear relationship with the annotation classes. Confirmed relevant documents have a substantially higher median score than the other classes. The median score is 0.514 for `confirmed_relevant`, 0.184 for `needs_review`, and 0.033 for `not_relevant`. This indicates that the score captures useful relevance signals. However, the distributions also overlap. Some non relevant documents receive high scores because they contain strong infrastructure terminology without confirming a TLM geometry update. Conversely, some relevant documents receive low scores when the decisive geometry signal is indirect, rare, or appears in a formulation not well covered by the scoring rules.

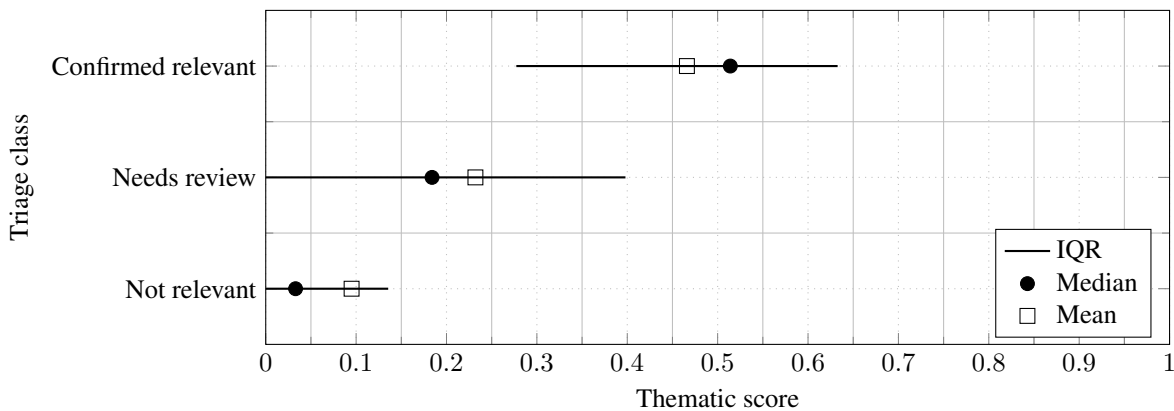


Figure 3: Thematic score distribution by triage class. The horizontal line shows the interquartile range, the filled circle shows the median, and the open square shows the mean. Confirmed relevant documents receive clearly higher thematic scores on average, but the overlap between classes shows that score based ranking alone cannot resolve all review decisions.

Overall, the exploratory analysis supports the chosen evaluation design. The dataset contains multiple source styles, the document lengths vary across classes, and the deterministic score provides a useful but imperfect ranking signal. These observations motivate the comparison of deterministic scoring with classical NLP, LLM based methods, and hybrid lead selection strategies in the following sections.

4 Methods

This section describes the methods used to prioritize official web documents for human review. The comparison follows the practical goal of ChangeScout: the output should not be an autonomous relevance decision, but a ranked review queue. The methods are therefore evaluated both as individual ranking or classification approaches and as components of a hybrid review workflow. This distinction matters because a false positive mainly increases review effort, while a false negative may hide a relevant or review worthy source.

4.1 Task Definition and Metrics

The evaluation uses the frozen datasets introduced in Table 4. Three related tasks are evaluated because the annotation schema separates confirmed relevance from review worthy uncertainty. The `strict_binary` task measures whether a method can distinguish confirmed TLM relevance from non relevance. The `needs_review` class is excluded from this task because these documents do not contain enough textual evidence for a confirmed relevance decision. The `actionable_binary` task measures whether a source should enter a human review queue. In this setting, both `confirmed_relevant` and `needs_review` are treated as positive. The `triage_3class` task keeps the full annotation structure and evaluates whether a method can distinguish confirmed relevance, review worthy uncertainty, and non relevance.

For the binary tasks, precision, recall, F1 score, and confusion matrices are reported. Continuous scores and probabilities are converted into binary predictions using thresholds selected on the corresponding train split. The test split is only used for final reporting. LLM triage outputs are mapped to the binary targets according to the same label definitions used for the evaluation datasets.

For lead selection, the evaluation additionally uses precision at N, recall at N, false negative count, and review workload reduction. These ranking oriented metrics are more appropriate than accuracy alone because ChangeScout is designed as a lead prioritization system. A useful method should surface many review worthy sources early and keep the number of missed relevant or uncertain cases low.

4.2 Ranking and Classification Methods

The first method is the deterministic thematic score. It assigns each cleaned document a continuous score between 0 and 1 based on configured infrastructure signals. The score is not treated as a final relevance decision, but as a transparent ranking signal for prioritizing documents before human review. It combines a rule based component and a retrieval component. The rule based component uses keyword and pattern groups for structural signals such as new roads, changed alignments, junctions, roundabouts, entries, exits, bridges, tunnels, underpasses, separated cycling or pedestrian infrastructure, and mapped road related objects. It also includes soft or negative signals that often occur in non relevant contexts, such as maintenance, resurfacing, temporary traffic management, markings, drainage, lighting, or administrative information.

The retrieval component uses BM25, a standard ranking function in information retrieval [5]. BM25 scores are normalized within a run and combined with the rule based score. In the evaluated configuration, the final thematic score is computed as a weighted combination of both components:

$$\text{thematic_score} = w_r \cdot \text{rule_score} + w_b \cdot \text{bm25_score} \quad (1)$$

where w_r and w_b are the configured weights for the rule based and BM25 components. This deterministic method is useful because it is transparent, reproducible, and independent of labelled training data. Its expected weakness is that it can overrate documents with many infrastructure terms but no confirmed geometry change, and underrate documents where the relevant geometry signal is indirect or expressed with uncommon wording.

The second method is a classical supervised NLP model using TF IDF features and Logistic Regression. TF IDF represents documents by weighting terms according to their frequency in a document and their specificity across the corpus [6]. Logistic Regression is then trained as a linear classifier on the frozen train split. This method was chosen because it is suitable for small text classification datasets, provides a learned signal, and remains computationally lightweight compared with larger neural models. It also serves as an important reference point for evaluating whether LLM based methods add value beyond a conventional NLP classifier.

The input text is built from the cleaned document title and body text. The model is evaluated on the same fixed splits as the deterministic score method. For the actionable binary task, the positive class consists of both `confirmed_relevant`

and `needs_review`. The classifier outputs a binary prediction and a positive class probability, which is later reused as one signal in hybrid lead selection. The implementation uses scikit learn [7].

The third method family consists of LLM based triage. These methods are evaluated to test whether instruction tuned language models can apply the annotation logic more flexibly than deterministic scoring or a classical sparse text classifier. The evaluated models include Qwen2.5 and Llama 3.1 variants [8, 9]. The models are not fine tuned for this project. Given the small size of the annotation dataset, fine tuning a neural classifier would not provide a robust basis for comparison. Instead, the LLMs are evaluated with controlled prompts that encode the frozen annotation semantics.

The prompt asks the model to classify each source into the same three classes used in the annotation schema: `confirmed_relevant`, `needs_review`, and `not_relevant`. The output is required as structured JSON containing the triage class, boolean label fields, a change type, notes, and evidence. Structured output is used because the results must be parsed and compared reproducibly. Two prompting strategies are evaluated. The direct prompt asks the model to assign a triage class directly. The hierarchical prompt mirrors the annotation logic more closely: it first checks whether the source contains a plausible TLM geometry signal and then checks whether the source provides enough evidence for confirmed relevance.

This prompt design follows the practical review problem. The hardest cases are often not clearly relevant or irrelevant. Many sources contain plausible signals but require additional information, such as plans, maps, project documents, or later updates, before a final binary decision can be made. LLM predictions are therefore not used as hard exclusion signals. An LLM prediction of `not_relevant` may be useful as a review signal, but it should not automatically remove a document selected by deterministic scoring or TF IDF probability.

4.3 Hybrid Lead Selection and Review Support

Hybrid lead selection evaluates practical strategies for building a ranked review queue. The positive target is actionable relevance, meaning that both `confirmed_relevant` and `needs_review` count as review worthy. This corresponds to the practical goal of ChangeScout: preserve relevant and uncertain sources while reducing the amount of irrelevant material a reviewer needs to inspect.

Several selection modes are compared. The `score_only` mode ranks documents by deterministic thematic score. The `tfidf_only` mode ranks documents by the TF IDF actionable probability. The `llm_only` mode ranks documents using the LLM triage signal. The `score_or_tfidf` mode selects documents if either the deterministic score or the TF IDF probability crosses the configured threshold. Additional hybrid modes combine score, TF IDF probability, and LLM triage information with recall oriented weighting or boost rules. The lead selection evaluation uses precision at N, recall at N, false negative count, and review workload reduction because the output is a ranked queue, not only a binary classification table.

After candidate leads have been selected, optional review support fields can be added to make manual inspection easier. This step does not decide whether a document enters the review queue and does not affect the relevance evaluation. The LLM explanation prompt can produce an `evidence_type`, an `explanation_note`, an `evidence_snippet`, a `geometry_signal`, and an `audit_warning`. These fields are intended to summarize why a document may be relevant and whether the text contains confirmed geometry evidence, only a plausible review signal, or no usable geometry evidence. In addition, deterministic local location hinting and optional GeoAdmin enrichment can provide possible place names or coordinates for selected leads. The GeoAdmin enrichment uses the GeoAdmin Search API to retrieve possible location candidates from Swiss federal geodata services [10]. These hints are review aids only. They do not verify the project location, do not confirm TLM relevance, and do not affect lead selection. This support layer is therefore treated as a proof of concept, while the evaluated NLP contribution remains the prioritization and triage of official texts.

5 Results

This section reports the performance of the individual methods, the hybrid lead selection strategies, and the optional review support layer. The interpretation follows the practical goal of ChangeScout: reducing manual screening effort while preserving confirmed relevant and review worthy sources.

5.1 Individual Method Performance

Table 7 shows the strict binary relevance results. This task excludes `needs_review` cases and evaluates confirmed TLM relevance only. On the frozen test split, the deterministic thematic score performs strongest, with an F1 score of 0.917. It reaches the same recall as TF IDF Logistic Regression, but with substantially higher precision. This shows that

the domain specific score is well aligned with clear textual evidence for confirmed geometry relevance in the evaluated source mix. The result should still be interpreted as task specific: the score is transparent and useful, but it is calibrated to the current source scope and should be rechecked for new source types.

Table 7: Strict binary relevance results. The `needs_review` class is excluded. Best values are shown in bold.

Method	Precision	Recall	F1	TP	FP	FN
Thematic score	0.957	0.880	0.917	22	1	3
TF IDF Logistic Regression	0.688	0.880	0.772	22	10	3
Qwen2.5 7B direct	1.000	0.600	0.750	15	0	10
Qwen2.5 7B hierarchical	1.000	0.640	0.780	16	0	9
Llama 3.1 8B hierarchical	0.929	0.520	0.667	13	1	12
Qwen2.5 14B direct	1.000	0.120	0.214	3	0	22
Qwen2.5 14B hierarchical	1.000	0.120	0.214	3	0	22

Table 8 shows the actionable binary results. This task is closer to the intended review workflow because both `confirmed_relevant` and `needs_review` count as positive. Qwen2.5 14B hierarchical achieves the highest F1 score with 0.838, mainly because it combines very high precision with moderate recall. TF IDF Logistic Regression reaches the highest recall among the individual methods, with 0.881. This makes it more useful as a high recall candidate selection signal. The thematic score remains competitive, but produces more false positives than the LLM based methods and slightly more false negatives than TF IDF.

Table 8: Actionable binary results. `confirmed_relevant` and `needs_review` are treated as positive. Best values are shown in bold.

Method	Precision	Recall	F1	TP	FP	FN
Thematic score	0.750	0.857	0.800	36	12	6
TF IDF Logistic Regression	0.771	0.881	0.822	37	11	5
Qwen2.5 7B direct	1.000	0.619	0.765	26	0	16
Qwen2.5 7B hierarchical	1.000	0.548	0.708	23	0	19
Llama 3.1 8B hierarchical	0.923	0.571	0.706	24	2	18
Qwen2.5 14B direct	0.938	0.714	0.811	30	2	12
Qwen2.5 14B hierarchical	0.969	0.738	0.838	31	1	11

The three class triage task is more difficult because it requires methods to separate confirmed relevance, review worthy uncertainty, and non relevance. Table 9 summarizes the LLM prompt comparison. Qwen2.5 7B hierarchical reaches the highest triage accuracy, while Qwen2.5 14B hierarchical reaches the highest actionable F1. Llama 3.1 8B was included as an additional model comparison point. Its results show that the observed triage behavior is not specific to the Qwen model family, although the experiment is too limited to draw broader conclusions about model architectures.

Table 9: LLM prompt comparison on the frozen triage test records. Best values are shown in bold.

Model	Prompt	Strict F1	Actionable F1	Triage accuracy
Qwen2.5 7B	direct	0.750	0.765	0.657
Qwen2.5 7B	hierarchical	0.780	0.708	0.671
Llama 3.1 8B	hierarchical	0.667	0.706	0.586
Qwen2.5 14B	direct	0.214	0.811	0.529
Qwen2.5 14B	hierarchical	0.214	0.838	0.571

Overall, no single method dominates all tasks. The thematic score is strongest for strict confirmed relevance. TF IDF Logistic Regression provides the strongest recall for actionable review selection. LLM based methods provide high precision and structured triage output, but they are too conservative to be used as standalone lead discovery methods.

5.2 Hybrid Lead Selection

Standalone metrics do not fully answer the practical question of ChangeScout. In practice, the output is a ranked review queue. The relevant question is how many confirmed relevant or review worthy sources appear within the first

N documents that a reviewer inspects. This value N is the review depth. Workload reduction is measured within the frozen test set. For example, reviewing 50 of 70 test documents means that 20 test documents are not reviewed, which corresponds to a workload reduction of 28.6%. This does not describe total production workload, because source maintenance, crawling, validation, and integration into production workflows are outside this metric.

Table 10 compares the main selection modes at $N = 50$. This review depth is useful because it still reduces the test review workload, while being large enough to test whether the ranking preserves most actionable sources. The `score_only` mode uses only the deterministic thematic score. The `tfidf_only` mode uses only the learned TF IDF actionable probability. The `score_or_tfidf` mode is a union strategy: a document can enter the review queue if either the deterministic thematic score or the TF IDF probability provides a strong enough signal. The LLM based hybrid variants additionally use LLM triage information for ranking or prioritization, but not as a hard exclusion signal.

Table 10: Lead selection comparison at review depth $N = 50$. The target is actionable relevance.

Selection mode	Precision@50	Recall@50	False negatives	Workload reduction
<code>score_only</code>	0.760	0.905	4	0.286
<code>tfidf_only</code>	0.740	0.881	5	0.286
<code>score_or_tfidf</code>	0.780	0.929	3	0.286
LLM hybrid variants	0.780	0.929	3	0.286

At $N = 50$, `score_or_tfidf` recovers 39 of 42 actionable sources. This is better than `score_only` and `tfidf_only`. The LLM based hybrid variants reach the same aggregate metrics at this review depth, but do not improve precision, recall, or false negatives. The simpler `score_or_tfidf` strategy is therefore preferred for high recall candidate selection, while LLMs are more useful after selection as explanation and review support.

The behavior at other review depths confirms this interpretation. At $N = 10$, several modes reach perfect precision but recover only 10 of 42 actionable sources. At $N = 20$, LLM based and hybrid variants still reach perfect precision, but recover less than half of all actionable sources. At $N = 70$, all modes reach full recall because the full test set is reviewed, so there is no workload reduction. The useful operating range is therefore between very shallow review queues, which miss too much, and full review, which saves no work.

Table 11 lists the three actionable records missed by the recommended `score_or_tfidf` strategy at $N = 50$. All three are broad Solothurn media or government communication pages. Their relevant TLM signal is present, but not prominent enough for the current thematic score, TF IDF probability, or LLM triage signal.

Table 11: False negatives of the recommended `score_or_tfidf` strategy at $N = 50$.

ID	Class	Short title	Source	Score	TF IDF
ann_0010	confirmed	Mobilitätsdrehscheibe Solothurn Süd	so_media	0.260	0.410
ann_0024	review	Regierungsratssitzung 28.10.2025	so_media	0.121	0.220
ann_0047	review	Regierungsratssitzung 29.10.2024	so_media	0.000	0.195

The false negatives support the earlier observation that media releases are useful hard negatives and that broadly written official communication remains difficult for automatic prioritization. This also supports the human in the loop design: LLM `not_relevant` predictions should not remove a lead when score or TF IDF signals are present.

5.3 Review Support Outputs

The optional explanation layer was evaluated on the Top 50 leads selected by `score_or_tfidf`. This is the same review queue used in the recommended high recall setup. The explanation model does not select or remove leads. It generates evidence fields for human review.

Table 12: LLM explanation support results for the Top 50 `score_or_tfidf` leads.

Measure	Value
Selected leads	50
Evidence snippets found in source	45
Missing evidence snippets	0
Manual explanation check flags	19
confirmed_geometry	22
plausible_review_signal	14
no_geometry_evidence	14

In 45 of 50 cases, the generated evidence snippet was found in the source text. This is useful in practice because reviewers receive a short evidence note and, where available, a source grounded snippet that indicates what should be checked in the original document. The evidence type distribution also shows that the high recall queue contains different kinds of leads: 22 with confirmed geometry evidence, 14 with plausible review signals, and 14 without usable geometry evidence.

The 19 manual check flags show that the explanation layer still needs control. It can make the review queue easier to use, but it does not replace source inspection. Its output should be treated as support for the reviewer, not as proof that a source is or is not TLM relevant.

6 Discussion

The results show that ChangeScout is best understood as a lead prioritization workflow, not as an autonomous classifier. This distinction is central to the project. The task is not to decide automatically whether `swissTLM3D` must be updated. The task is to prioritize official web texts so that human reviewers can inspect the most relevant sources earlier and with less manual screening effort. Seen from this perspective, the most useful setup is not one single method, but a combination of deterministic thematic scoring, TF IDF based candidate selection, and optional LLM based review support.

6.1 Interpretation of the Method Comparison

The deterministic thematic score performed strongly in the strict binary task, where only confirmed relevance and non relevance are compared. This is plausible because confirmed TLM relevance often depends on explicit domain signals such as new roads, changed alignments, new roundabouts, tunnels, underpasses, bridges, or separated cycling infrastructure. A transparent score that encodes such patterns can work well when official project texts use similar vocabulary. However, the result should be interpreted as task specific. The thematic score is an engineered domain heuristic shaped by the annotation guidelines, TLM relevance logic, and early project experience. It should therefore be revalidated when applied to procurement documents, PDF heavy sources, municipal pages, or other communication styles.

TF IDF Logistic Regression adds a complementary learned signal. In the actionable binary task, it achieved the highest recall among the individual methods, which is important because actionable relevance includes both confirmed relevant sources and sources that require further review. The model can learn recurring formulations from the labelled dataset, including terms that are not fully captured by hand written rules. For this project, TF IDF is therefore not only a comparison baseline, but part of the recommended candidate selection strategy.

The LLM based methods produced structured outputs and often high precision, but their recall was too low for standalone lead discovery. This conservative behavior is especially visible in strict relevance, where confirmed relevant sources were often downgraded instead of classified as confirmed relevant. LLM outputs are therefore risky as hard exclusion signals. Their stronger role is after selection: they can generate explanation notes, distinguish confirmed geometry evidence from weaker review signals, and help reviewers inspect selected leads more efficiently. The results support a clear division of labor: deterministic scoring and TF IDF for candidate selection, LLMs for explanation and review support.

6.2 Why the Review Queue Is the Right Evaluation Object

Standalone F1 scores are useful, but they do not fully answer the main project question. A reviewer does not receive a model metric. A reviewer receives a queue of documents. The relevant operational question is therefore how many

confirmed relevant or review worthy sources appear within the first documents that a human actually inspects. This is why Precision@N, Recall@N, false negatives after N, and workload reduction are more meaningful than accuracy alone.

The Top 50 `score_or_tfidf` setup illustrates this point. It recovered 39 of 42 actionable sources while reducing the test set review workload by 28.6%. It did not maximize precision, but it kept false negatives low. For this use case, that is the more useful tradeoff. A false positive costs review time. A false negative can hide a source that should have been checked. The result also explains why the simplest high recall strategy is preferable to more complex hybrid variants at this review depth: the LLM based hybrid variants reached the same aggregate metrics, but did not improve precision, recall, or false negatives beyond `score_or_tfidf`. If a simpler method reaches the same result without relying on LLM triage, it is the better candidate selection layer.

The `needs_review` class is important for the same reason. It is not merely label noise. It represents a real production situation: the source contains a plausible TLM signal, but the text alone does not provide enough evidence for a final decision. These documents belong in the review queue because a human may need to inspect plans, maps, attachments, later project updates, or other context. Treating them as irrelevant would make the evaluation cleaner mathematically, but less faithful to the actual review workflow.

6.3 Failure Modes and Limits of the Current Evaluation

The main difficulty is that infrastructure language does not map directly to TLM relevance. Some documents contain many construction terms but no persistent geometry change. Other documents contain a relevant signal, but the signal is weak, indirect, or hidden inside broad administrative communication. This explains both false positives and false negatives. Maintenance, resurfacing, drainage, temporary traffic management, and general planning concepts can look relevant to a text model even when no mapped geometry would change. Conversely, small geometry changes, traffic islands, or relevant details inside government meeting summaries can be missed because the decisive signal is not prominent.

Table 13: Qualitative failure modes observed during error analysis.

Failure mode	Typical cause	Effect
Maintenance overcalled	Rehabilitation, resurfacing, repair, drainage	False positives
Planning text overcalled	Concepts without confirmed geometry evidence	False positives or review cases
Cycling infrastructure ambiguity	Painted cycling lane versus separated cycling path	Wrong relevance decision
Bridge replacement overinterpreted	Replacement without mapped geometry change	False positives
Small mapped objects missed	Weakly expressed traffic islands or local changes	False negatives
Temporary traffic misread	Construction traffic management	False positives
Hidden signal missed	Relevant detail in broad communication	False negatives

The false negatives of the recommended Top 50 setup are a good example of this limitation. All three missed actionable cases are broad Solothurn media or government communication pages. Their relevant signal is present, but not prominent enough for the current thematic score, TF IDF probability, or LLM triage signal. This confirms that media releases are useful hard negatives, but also difficult sources. They contain many official topics, only some infrastructure content, and only occasional TLM relevant signals. This is exactly the type of source where automatic exclusion would be risky.

The evaluation remains limited by dataset size and source scope. The full annotation dataset contains 348 documents, and the aligned test set contains 70 records. This is enough to compare methods within the project, but not enough for broad claims about all public infrastructure communication in Switzerland. The dataset covers selected official cantonal web pages and a deliberate media sample. It does not cover the full complexity of procurement platforms, PDFs, municipal websites, APIs, or structured data exchanges. Performance differences should therefore be interpreted cautiously when they are small. The strongest conclusions are those supported by several signals at once, such as TF IDF improving actionable recall and `score_or_tfidf` reducing false negatives at useful review depth.

There is also an important conceptual limitation. ChangeScout evaluates textual leads, not actual TLM updates. A lead means that a source should be reviewed. It does not prove that the landscape has already changed, that the change is visible, or that a specific TLM object must be updated. Location hints and GeoAdmin enrichment can help reviewers orient themselves, but they do not verify project geometry. The same applies to LLM explanations: they can make a lead easier to inspect, but they are not evidence that an update is required.

6.4 Implications for Use

The results support a human in the loop use of ChangeScout. Candidate selection should rely on `score_or_tfidf`, because this combines transparent domain knowledge with a learned TF IDF signal and gives the best high recall tradeoff at useful review depth. This setup is preferable to an LLM based hard filter, because LLM false negatives remain too risky for lead discovery. LLMs should be used after candidate selection. Their useful role is evidence generation, explanation notes, triage support, and possibly priority support. An LLM `not_relevant` prediction should not remove a lead when the thematic score or TF IDF probability is strong. Location hints should be treated in the same way: useful for orientation, but not as verified geometries. The practical implication is that ChangeScout should support the existing review workflow. It can make public textual evidence easier to find, rank, and inspect, while final judgement remains with domain experts.

7 Conclusion and Outlook

This project investigated whether NLP methods can prioritize official web texts so that TLM relevant infrastructure changes can be reviewed more efficiently by humans. The results show that this is feasible. Official cantonal web pages can contain useful textual signals for topographic data production, and these signals can be turned into a ranked review queue. ChangeScout should therefore be understood as a review support system, not as an automatic TLM update mechanism.

The deterministic thematic score performed strongly for confirmed relevance, which shows that transparent domain rules are valuable in this task. TF IDF Logistic Regression added an important learned signal and achieved the strongest recall for actionable review selection among the individual methods. LLM based methods were less suitable as standalone lead discovery methods because of conservative recall behavior, but they added value as explanation and review support components. The best practical setup is therefore a hybrid workflow: first combine deterministic thematic scoring with TF IDF based actionable relevance probabilities to select candidates with high recall, then use LLM based explanation and location hints to support human review.

The main limitation is that the evaluation is still based on a controlled source scope and a relatively small annotated dataset. The results should therefore not be interpreted as a production performance guarantee for all Swiss public infrastructure sources. More source types, more annotated data, and real reviewer feedback are needed before the workflow can be used operationally. Future work should extend the source coverage beyond the current cantonal web pages. This includes additional cantons, municipal sources, procurement platforms, PDFs, project plans, and structured data exchanges. PDF and plan extraction are especially important, because many relevant infrastructure details are not fully described in public web descriptions. The system should also include active learning and reviewer feedback, so that manual decisions can improve the annotation dataset and the TF IDF model over time.

A further development step is stronger geographic linking. The current location hints and GeoAdmin enrichment are useful review aids, but they do not verify project geometry. Future work should evaluate geoparsing, disambiguation, and linking of textual leads to GIS objects or project areas. Finally, an operational version would need temporal monitoring, run level quality checks, source drift monitoring, and clear integration into the existing production review process. With these extensions, ChangeScout could become a practical component for shortening the reaction time between real world infrastructure changes and topographic data production.

Acknowledgements

I thank the Federal Office of Topography swisstopo for supporting my participation in the CAS Natural Language Processing program and for providing the professional context in which this project was developed. I also thank the CAS Natural Language Processing teaching team at the University of Bern for the course content, exercises, and learning environment that supported this project. Computational resources for local model evaluation were provided by the UBELIX HPC cluster of the University of Bern. AI assistance was used throughout the project for coding support, debugging, structuring, language editing, and wording refinement. The substantive project work, including the research framing, domain interpretation, annotation logic, evaluation choices, and final assessment of the results, was carried out by the author.

References

- [1] Federal Office of Topography swisstopo. swissTLM3D: The Large Scale Topographic Landscape Model of Switzerland. <https://www.swisstopo.admin.ch/de/landschaftsmodell-swisstlm3d>, 2026. Accessed:

2026-05-11.

- [2] Lorenz Joss. ChangeScout Source Registries. <https://github.com/loloj-0/changescout/tree/main/config/sources>, 2026. Versioned YAML source registry files. Accessed: 2026-05-11.
- [3] Lorenz Joss. ChangeScout Project Repository. <https://github.com/loloj-0/changescout>, 2026. Accessed: 2026-05-11.
- [4] Lorenz Joss. ChangeScout Annotation Guidelines. https://github.com/loloj-0/changescout/blob/main/docs/annotation_guidelines.md, 2026. Accessed: 2026-05-11.
- [5] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.
- [6] Gerard Salton and Christopher Buckley. Term weighting approaches in automatic text retrieval. *Information Processing and Management*, 24(5):513–523, 1988.
- [7] Fabian Pedregosa, Gael Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Edouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [8] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.
- [9] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- [10] Federal Office of Topography swisstopo. GeoAdmin API Search Service Documentation. <https://docs.geo.admin.ch/access-data/search.html>, 2026. Accessed: 2026-05-11.